

- **Example** An object with mass m that moves in a circular path with constant angular speed ω has position vector $\mathbf{r}(t) = a \cos \omega t \mathbf{i} + a \sin \omega t \mathbf{j}$. Find the velocity and acceleration of the object. Also, find the force acting on the object.

Newton's Second Law of Motion: $\mathbf{F}(t) = m\mathbf{a}(t)$

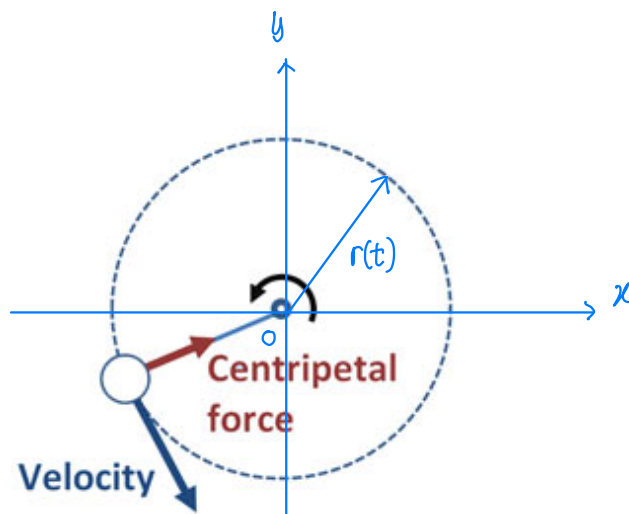
$$\vec{r}(t) = \langle a \cos \omega t, a \sin \omega t \rangle$$

$$\vec{v}(t) = \frac{d}{dt} \vec{r}(t) = \langle -a\omega \sin \omega t, a\omega \cos \omega t \rangle$$

$$\vec{a}(t) = \frac{d^2}{dt^2} \vec{r}(t)$$

$$= \frac{d}{dt} \vec{v}(t)$$

$$= \langle -a\omega^2 \cos \omega t, -a\omega^2 \sin \omega t \rangle$$



$$\vec{F}(t) = m \vec{a}(t)$$

$$= \langle -maw^2 \cos wt, -maw^2 \sin wt \rangle$$

$\vec{r}(t)$
 "

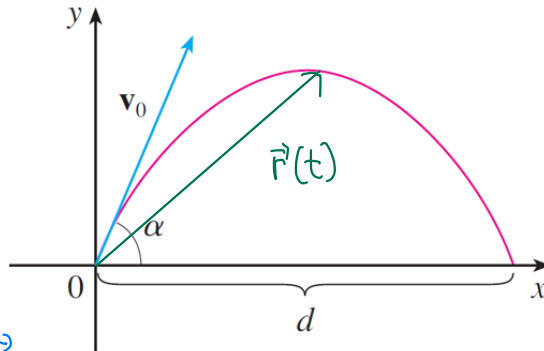
$$= -m\omega^2 \boxed{\langle a \cos wt, a \sin wt \rangle}$$

$$= -m\omega^2 \vec{r}(t)$$

↑

force acts in the direction opposite to the position vector $\vec{r}(t)$ and therefore points toward the origin.

- **Example** A projectile is shot from the origin at an angle α from the positive x -axis with an initial velocity of $\mathbf{v}_0$. Assuming that air resistance is negligible and the only external force is due to gravity ($g \approx 9.8 \text{ m/s}^2$), find the position function $\mathbf{r}(t)$ of the projectile. What value of α maximizes the range, d ?



$$\vec{a}(t) = -g \vec{j}$$

$$\vec{v}(t) = c_1 \vec{i} + (c_2 - gt) \vec{j}$$

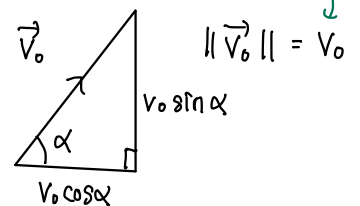
$$\vec{v}(0) = c_1 \vec{i} + c_2 \vec{j} = \vec{v}_0$$

$$\therefore \vec{v}(t) = \vec{v}_0 - gt \vec{j}$$

$$\vec{r}(t) = \vec{v}_0 t - \frac{1}{2}gt^2 \vec{j} + \cancel{\vec{r}_0} \rightarrow \vec{0} \text{ as } \vec{r}(0) = \vec{0} \quad \text{initial speed}$$

$$\therefore \vec{r}(t) = \vec{v}_0 t - \frac{1}{2}gt^2 \vec{j}$$

$$= \langle v_0(\cos \alpha)t, v_0(\sin \alpha)t - \frac{1}{2}gt^2 \rangle$$



The trajectory is a parabola with parametric equations $\vec{v}_0 = \langle v_0 \cos \alpha, v_0 \sin \alpha \rangle$

$$x = (v_0 \cos \alpha)t \quad y = (v_0 \sin \alpha)t - \frac{1}{2}gt^2$$

$$\vec{r}(t) = \langle v_0(\cos \alpha)t, v_0(\sin \alpha)t - \frac{1}{2}gt^2 \rangle$$

When $x = d$, $y = 0$.

Want t , when $\vec{r}(t) = \langle d, 0 \rangle$.

$$v_0(\sin \alpha)t - \frac{1}{2}gt^2 = 0$$

$$\Rightarrow t \left[v_0(\sin \alpha) - \frac{1}{2}gt \right] = 0$$

$$\Rightarrow t = 0,$$

(not suitable)

$$v_0(\sin \alpha) - \frac{1}{2}gt = 0$$

$$\Rightarrow t = \frac{2 v_0 \sin \alpha}{g}$$

So,

$$d = (v_0 \cos \alpha) t$$

$$= (v_0 \cos \alpha) \left[\frac{2 v_0 \sin \alpha}{g} \right]$$

$$= \frac{v_0^2 \sin(2\alpha)}{g}$$

d is maximized when $\sin(2\alpha) = 1$

$$\Rightarrow 2\alpha = \frac{\pi}{2}$$

$$\Rightarrow \alpha = \frac{\pi}{4}$$

v_0

- **Example** A projectile is fired with muzzle speed 150 m/s and angle of elevation 45° from a position 10 m above ground level. Where does the projectile hit the ground, and with what speed?

$$x = (v_0 \cos \alpha)t \quad y = (v_0 \sin \alpha)t - \frac{1}{2}gt^2 + 10$$

$$x = 150 (\cos 45^\circ) t$$

$$y = 150 (\sin 45^\circ) t - \frac{1}{2} (9.8) t^2 + 10$$

Hit the ground $\Rightarrow y = 0$:

$$150 (\sin 45^\circ) t - \frac{1}{2} (9.8) t^2 + 10 = 0$$

$$\Rightarrow 4.9 t^2 - 75\sqrt{2} t - 10 = 0$$

$$\Rightarrow t = \frac{75\sqrt{2} \pm \sqrt{11250 + 196}}{9.8}$$

$$= 21.74 \text{ s} \quad \text{or} \quad -0.09 \quad (\text{not suitable, as } t \geq 0)$$

Where ? $x = 150 (\cos 45^\circ) (21.74)$
 $= 2306 \text{ m}$

Speed ?

$$\vec{r}(t) = \left\langle 150 (\cos 45^\circ) t, 150 (\sin 45^\circ) t - \frac{1}{2} (9.8) t^2 + 10 \right\rangle$$

$$\vec{v}(t) = \frac{d}{dt} \vec{r}(t)$$

$$= \left\langle 150 (\cos 45^\circ), 150 (\sin 45^\circ) - 9.8t \right\rangle$$

$$\vec{v}(21.74) = \left\langle 75\sqrt{2}, 75\sqrt{2} - 213.052 \right\rangle$$

speed

$$v = \sqrt{(75\sqrt{2})^2 + (75\sqrt{2} - 213.052)^2}$$

$$= 151 \text{ m/s}$$